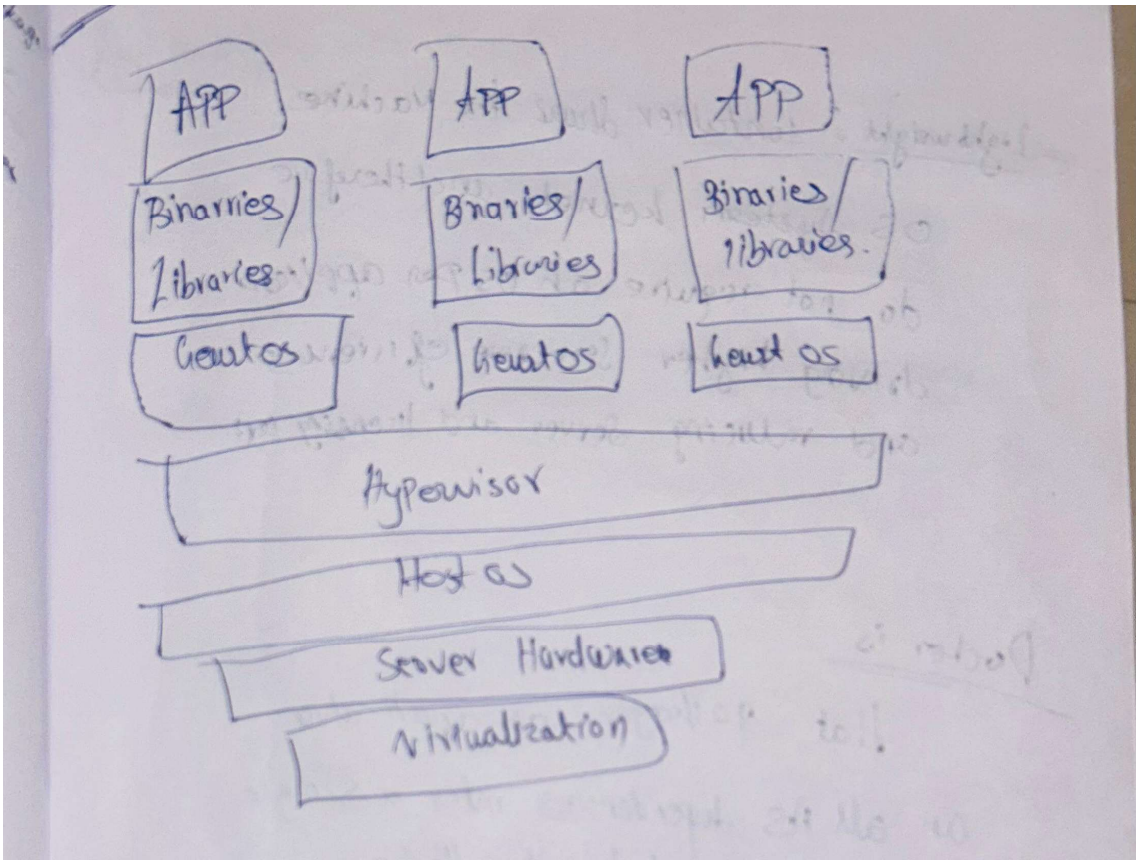
ASSIGNMENT 16

DOCKER INTRODUCTION THEORY

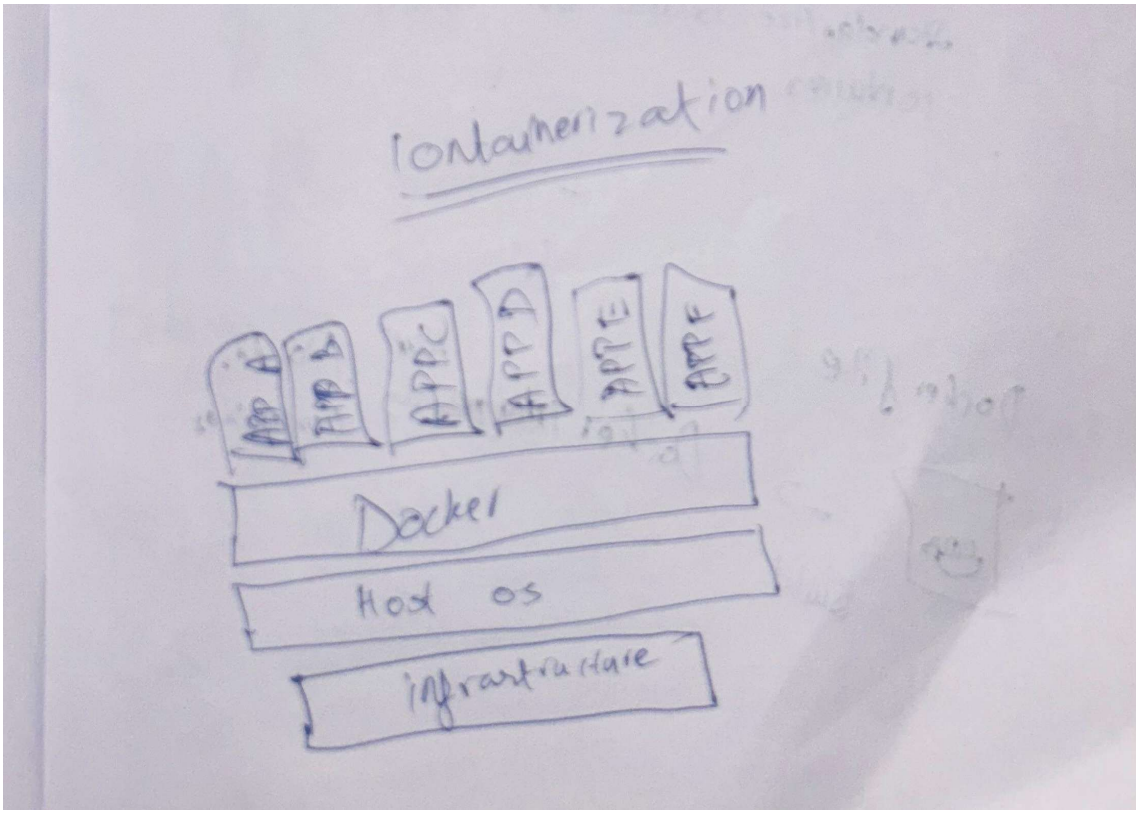
VIRTUALIZATION :

Virtualization means running multiple virtual computers on one physical computer.
It uses software called a **hypervisor** to share hardware resources.
Each virtual machine works like a real computer.
It saves cost, space, and power.
It is widely used in cloud computing.



CONTAINERIZATION :

Containerization means running applications in isolated containers on one system.
Containers share the same operating system kernel.
They are lightweight and start very fast.
It makes applications easy to move and deploy.
It is widely used with Docker and Kubernetes.



Lightweight : container share the Machine OS system kernel and therefore do not require an OS per application driving higher server efficiencies and reducing server and licensing cost.

Docker is

that packages an application with all its dependencies into a single standardize isolated unit called a container.

Docker file → Build → Docker image → Run → Docker container

KEY COMPONENTS OF DOCKER :

Key components of Docker

Docker Engine: The runtime that executes and manages container

Docker images: Read only templates that define the contents of a container

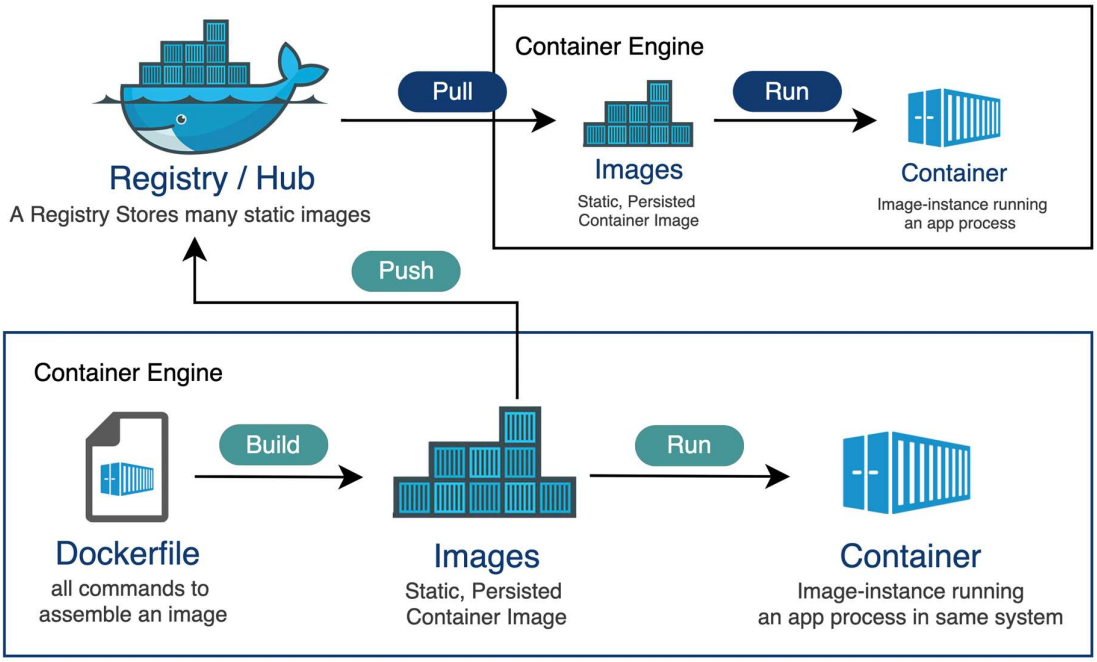
Docker container: Instance of Docker images running as isolated.

Docker Hub: Public Registry for sharing and managing Docker images.

DOCKER REGISTRY :

Docker registry :-

Docker Registry is a centralized storage and distribution system for Docker images which are building blocks for running application in containers. it is like a library or package manager, allowing user to upload (Push) and download (Pull) container



Docker Registry is a place to store Docker images.
It allows users to upload and download images.
Docker Hub is a popular public registry.
Private registries are used inside companies.
It helps share and manage container images easily.